ECE750T: Data Documentation Audit

Florence-2-base

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Fall 2026

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1. Context [8 marks]

I audit Microsoft's Hugging Face model card for `microsoft/Florence-2-base`, a 230-million-parameter vision model. The artifact is the repository README at `https://huggingface.co/microsoft/Florence-2-base`, accessed October 14, 2026, revision `5ca5edf5` (full revision in the references). Its linked technical report supplies supporting evidence; I distinguish information in that report from information in the card [Microsoft, n.d., Xiao et al., 2023].

The proposed use is helping a university library create searchable records for digitized historical posters and flyers. Staff would review literal transcriptions and draft descriptions of illustrations before indexing them, retaining the scans as the sources. This is my assessment scenario. The card describes general visual tasks, including captioning, detection and optical character recognition (OCR), without establishing this archival use. Its examples and benchmarks suggest an audience of developers and researchers. It explains capabilities and how to use them; this audit asks whether it also lets librarians judge suitability. I apply the Model Cards framework [Mitchell et al., 2019].

2. Critique [20 marks]

Strengths. The card's "Model Summary" distinguishes base and fine-tuned checkpoints. Under "Tasks," OCR returns text, while OCR with region returns text and coordinates; captioning is a separate task. These distinctions help a developer select outputs and potentially link recognized text to an image region. Named benchmarks and the linked paper provide evidence beyond demonstrations. Results for `base-ft` must not be attributed to the base model.

Archival use boundaries are missing. The card does not explicitly define primary users, out-of-scope uses or deployment caveats, as recommended by Mitchell et al. [2019, §§4.2, 4.8–4.9]. For this use, a transcription should preserve visible wording, while a description interprets visual content. The documentation gives no archival guidance on illegible text, historical spelling or separating those fields. A plausible completion of a damaged name could become a false catalogue fact; an incorrect transcription could hide an item from a name search. These are possible failure pathways, not observed Florence-2 errors in this audit.

The metrics do not establish reliable discovery. The card reports 70.1 CIDEr for base-model captioning on TextCaps. CIDEr measures agreement with reference descriptions [Vedantam et al., 2015]; it is not a character-transcription accuracy measure. The card does not report character or word error rates for historical posters, nor retrieval results for their names and dates. The intended construct is faithful, useful access to archival material. Treating caption similarity as sufficient evidence would leave a construct-validity gap [Jacobs and Wallach, 2021, §3.2]. Even low average transcription error could conceal a wrong name that determines whether a record is found. The Library of Congress describes how OCR supports searching while unusual text styles and small fonts can produce errors [Library of Congress, n.d.]; this motivates testing such conditions, without establishing Florence-2's performance on them.

Coverage and uncertainty are unresolved. The Model Cards framework calls for relevant factors and disaggregated evaluation [Mitchell et al., 2019, §§4.3, 4.7]. The paper identifies source datasets and automated annotation, including an OCR service (§§4.1–4.3), so training provenance is not entirely absent. However, neither the card nor that account quantifies coverage or transcription accuracy by historical period, language, script, typeface, damage or layout. If errors concentrate in particular languages, those communities' records could become less discoverable. Missing evidence establishes uncertainty, not demonstrated discrimination. The card also lacks uncertainty estimates and a complete benchmark reproduction recipe; its examples do not establish that staff review will reliably correct errors.

3. Proposed changes [12 marks]

First, document the archival workflow and its limits. Microsoft maintainers should state which transcription and captioning uses have been evaluated. Librarians should define transcription conventions, review responsibilities and publication criteria. Add guidance to mark illegible spans, preserve historical spelling, and keep literal text separate from generated descriptions and inferred catalogue metadata. Retain links to scans and distinguish machine drafts from staff corrections. This would help readers judge which fields need verification without inventing model capabilities.

Second, supply evaluation evidence for this collection. Library staff and language specialists should create checked reference transcriptions from a representative sample, recording uncertainty in genuinely unreadable spans. Independent evaluators could compare Florence-assisted review with manual transcription and an established OCR workflow. Report character and word error rates, exact transcription of names and dates, omitted text regions, unsupported descriptive claims, retrieval recall for agreed queries, and correction time. Define scoring conventions and test reading order; report results by period, language/script, typeface, scan quality and damage, with sample counts and uncertainty. Split related editions or near-duplicate posters together to avoid leakage. Librarians and researchers should set acceptance criteria before testing; untested groups remain unresolved. These are proposed evaluations, not completed results.

Third, make coverage and results traceable. Microsoft's data team could provide source proportions, language/script coverage, annotation filters and human checks of automated labels, identifying unknowns. Maintainers and evaluators should record checkpoint and documentation revisions, test splits, image preprocessing, prompts, decoding settings, output limits and scoring code. Describe how truncation is detected and how corrections are logged; provide a changelog and feedback contact. This would let a library assess whether published evidence transfers to its material and monitor corrections. Current documentation supports planning a supervised trial, but does not establish dependable archival transcription or search.

References

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AI-assistance disclosure

AI assistance is fully permitted under the updated assignment policy. I used Codex (OpenAI) for source research, analysis, drafting and revision, LaTeX preparation, and generation of the PDF and section screenshots. The final interpretation and submission remain my responsibility. No model inference experiment or archival user study was conducted; all proposed evaluations are future work.